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Module 2 Machine Learning Overview

Module 2.1: Machine Learning Algorithms — 20 Questions

Single Choice Questions (1-7)

Q1 : A Huawei autonomous driving team is developing a system to detect traffic signs. The system is trained on 50,000 labeled images of traffic signs (stop signs, speed limits, yield signs). During testing, the system correctly identifies 47,300 signs out of 50,000 test images.
According to the machine learning framework (E, T, P), which statement correctly identifies all three components?
A. E = 50,000 training images; T = Detecting traffic signs; P = 94.6% accuracy
B. E = 47,300 correct predictions; T = Image classification; P = Training time
C. E = Labeled image dataset; T = Autonomous driving; P = Number of epochs
D. E = Testing phase; T = Traffic sign recognition; P = 50,000 test images

Q2 In the context of the Go game learning example from the material, a system first learns by playing millions of games against itself without any human intervention (Experience E_1). Then, it is fine-tuned using human expert game records (Experience E_3).
This training approach best represents the evolution from:

A. Unsupervised learning → Supervised learning
B. Reinforcement learning → Supervised learning
C. Semi-supervised learning → Supervised learning
D. Unsupervised learning → Semi-supervised learning

Q3 : financial institution wants to build a fraud detection system. The characteristics are:
- **Fraud patterns change constantly as criminals adapt**
- **Rules are extremely complex and interdependent**
- **Data volume is massive (millions of transactions daily)**
- **The system must adapt to new fraud patterns automatically**

Based on the "When to Use Machine Learning" matrix, which quadrant does this problem belong to?

A. Simple questions (Low complexity, Small scale)
B. Manual rules (High complexity, Small scale)
C. Rule-based algorithms (Low complexity, Large scale)
D. Machine learning algorithms (High complexity, Large scale)

Q4 The material states: "The objective function f is unknown, and the learning algorithm cannot obtain a perfect function f. Hypothesis function g approximates function f, but may be different from function f."
In a linear regression scenario where the true relationship is $y = 2x^2 + 3x + 1$, but the model assumes $y = wx + b$, what is the fundamental issue?

A. High variance (overfitting)
B. High bias (underfitting) due to model misspecification
C. Insufficient training data
D. Perfect approximation since $g \approx f$

Q5 A Huawei Cloud customer wants to predict the exact temperature for the next 7 days in Shenzhen based on historical weather data (temperature, humidity, pressure, wind speed). The output needs to be a continuous numerical value.
This is BEST classified as:

A. Classification task
B. Regression task
C. Clustering task
D. Reinforcement learning task

Q6 A company has 10 million customer transaction records with NO labels. They want to automatically group customers into segments based on purchasing behavior patterns, so they can create targeted marketing campaigns.
This task is:

A. Supervised classification
B. Unsupervised clustering
C. Supervised regression
D. Semi-supervised learning

Q7 In the traditional rule-based method vs. machine learning comparison diagram, a spam email filter is built using explicit if-then rules (e.g., "if email contains 'win free prize' → spam"). After 3 months, the filter's accuracy drops from 95% to 60% because spammers started using image-based text and obfuscated words.
According to the material, what is the PRIMARY reason rule-based methods failed?

A. The rules were not complex enough initially
B. The data distribution changed over time, and rules cannot self-adapt
C. The training dataset was too small
D. The performance measure P was incorrectly defined

Multiple Choice Questions (8-11) ⚠ (Select ALL that apply)

Q8 Which of the following scenarios from the material demonstrate situations where machine learning is PREFERRED over traditional rule-based methods? (Select all that apply)

A. Speech recognition where acoustic patterns are complex and variable
B. A simple calculator performing basic arithmetic operations
C. Part-of-speech tagging where new words and meanings emerge constantly
D. Sales trend forecasting where data patterns evolve over time
E. A traffic light system with fixed timing intervals

Q9 The material states: "Deep learning is a sub-field of machine learning. To understand deep learning, you need to first understand the fundamentals of machine learning."
Which of the following machine learning fundamentals are ESSENTIAL prerequisites for understanding deep learning? (Select all that apply)

A. Understanding of hypothesis functions and approximation
B. Knowledge of supervised vs. unsupervised learning paradigms
C. Experience with traditional rule-based programming only
D. Understanding of training data, validation, and model evaluation
E. Familiarity with manual rule extraction techniques

Q10 In the Go game learning example, three types of experience are described (E_1, E_2, E_3). Which statements correctly describe the characteristics of these experiences? (Select all that apply)

A. E_1 (Playing with itself) involves a feedback loop between the learning system and training process
B. E_2 (Inquiring humans) represents semi-supervised learning with partial human guidance
C. E_3 (Learning from historical games) is supervised learning with labeled historical data
D. All three experiences require a fixed dataset with pre-defined labels
E. E_1 is the only experience type that qualifies as true machine learning

Q11 A machine learning model for insurance claim prediction outputs a hypothesis function g that approximates the true objective function f. During deployment, the model consistently underestimates high-value claims by 30% but performs well on low-value claims.
Which statements about this scenario are correct? (Select all that apply)

A. The hypothesis function $g \neq f$, which is expected in machine learning
B. The model has acceptable performance since $g \approx f$ in the material's framework
C. The 30% systematic error on high-value claims indicates a significant approximation problem
D. This scenario demonstrates that the learning algorithm failed to obtain a perfect function
E. The model should be retrained only if the performance measure P is redefined

True or False Questions (12-15)

Q12 In the machine learning framework (E, T, P), the performance measure P must always be a single metric (e.g., accuracy) and cannot combine multiple evaluation criteria.

A. True
B. False

Q13 According to the material, reinforcement learning algorithms experience a fixed dataset during training, similar to supervised learning algorithms.

A. True
B. False

Q14 The diagram comparing traditional rule-based methods and machine learning shows that in rule-based methods, explicit programming is used and rules can be manually determined, while in machine learning, machines automatically learn rules that may be too complex for humans to explicitly describe.

A. True
B. False

Q15 Classification and regression are both prediction tasks, but classification outputs discrete class values while regression outputs continuous numerical values. Therefore, a model cannot perform both classification and regression simultaneously.

A. True
B. False

Fill in the Blank Questions (16-18)

Q16 In the machine learning framework, a computer program is said to learn from __________ (E) with respect to some class of __________ (T) and __________ (P), if its performance at tasks in T, as measured by P, improves with experience.

Q17 In the "When to Use Machine Learning" matrix, when the complexity of rules is __________ and the scale of the problem is __________, machine learning algorithms are the optimal choice compared to manual rules, simple questions, or rule-based algorithms.

Q18 According to the rationale diagram of machine learning algorithms, the learning algorithm takes __________ data D: $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ as input and produces a __________ function g that approximates the unknown __________ function f: X → Y.

Drag and Drop / Matching Questions (19-20)

Q19 Match each Go game learning experience with its correct learning type:

Experience	Learning Type
E_1 - Playing games with itself	A. Supervised learning
E_2 - Inquiring humans during self-play	B. Unsupervised learning
E_3 - Learning from human historical games	C. Semi-supervised learning
	D. Reinforcement learning

Write your answer as: E_1→?, E_2→?, E_3→?

Q20 Classify each of the following Huawei Cloud customer scenarios into the correct machine learning task type:

Scenario	Task Type
A. Predicting whether a customer will churn (leave) within 30 days	1. Classification
B. Grouping network logs into anomaly patterns without labels	2. Regression
C. Forecasting exact server CPU utilization for next 24 hours	3. Clustering
D. Categorizing support tickets into "Billing", "Technical", "Account"	

Write your answer as: A→?, B→?, C→?, D→?

Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	A	E=training data (experience), T=detecting signs (task), P=accuracy (performance)
2	B	E_1 is reinforcement learning (self-play with environment feedback), E_3 is supervised
3	D	High complexity (complex interdependent rules) + Large scale (millions of transactions)
4	B	Assuming linear model for quadratic data = model misspecification = high bias/underfitting
5	B	Continuous numerical output = Regression
6	B	No labels + grouping by similarity = Unsupervised clustering
7	B	Material explicitly states data distribution changes over time as a key ML use case
8	A, C, D	B and E are simple, fixed-rule scenarios unsuitable for ML
9	A, B, D	C and E are rule-based, not ML fundamentals needed for deep learning
10	A, B, C	D is false (E_1 doesn't need labels), E is false (all are ML experiences)
11	A, C, D	B is false (30% error is not acceptable approximation), E is false (retrain, don't redefine P)
12	B - False	P can be multi-metric (accuracy, precision, recall, F1, etc.)
13	B - False	RL interacts with environment, not fixed dataset (material explicitly states this)
14	A - True	Direct quote from the material's comparison diagram
15	B - False	Some models can do both (e.g., neural networks with different output layers)
16	experience, tasks, performance measure	Direct definition from material
17	High, Large	Top-right quadrant of the matrix
18	Training, hypothesis, target/objective	From the rationale diagram
19	$E_1 \to D$, $E_2 \to C$, $E_3 \to A$	E_1=Reinforcement, E_2=Semi-supervised, E_3=Supervised
20	A→1, B→3, C→2, D→1	Churn=binary classification, Logs=clustering, CPU=regression, Tickets=classification

Module 2.2: Types of Machine Learning — 20 Questions

Single Choice Questions (1-7)

Q1 A Huawei Cloud customer has a dataset of 1 million customer support tickets. Only 5,000 tickets are labeled with categories ("Billing", "Technical", "Account"). The customer wants to build a model to automatically categorize the remaining 995,000 tickets.
Which machine learning approach is MOST appropriate?

A. Pure supervised learning using only the 5,000 labeled tickets
B. Pure unsupervised learning ignoring all labels
C. Semi-supervised learning combining 5,000 labeled + 995,000 unlabeled tickets
D. Reinforcement learning with reward based on ticket resolution time

Q2 According to the material, reinforcement learning "can be understood as unsupervised learning plus supervised learning (unsupervised learning before supervised learning)."

In this interpretation, which phase corresponds to "unsupervised learning" and which to "supervised learning"?

A. Exploration phase = unsupervised; Exploitation phase = supervised
B. Environment interaction = unsupervised; Reward evaluation = supervised
C. Action generation without reward = unsupervised; Reward-based adjustment = supervised
D. Random action selection = unsupervised; Optimal policy learning = supervised

Q3 A stock prediction system predicts the exact price of Huawei stock for next Tuesday. The material classifies this as:

A. Supervised classification
B. Supervised regression
C. Unsupervised clustering
D. Reinforcement learning

Q4 In the reinforcement learning diagram (Model ↔ Environment), at time step t, the model:

1. **Receives state s_t from environment**
2. **Takes action a_t**
3. **Receives reward r_{t+1} and new state s_{t+1}**

Which statement about this feedback loop is CORRECT?

A. The reward r_{t+1} is provided by a human supervisor telling the model the correct action
B. The environment provides scalar reinforcement signals to evaluate actions, not correct actions
C. The model knows the optimal action before taking it
D. The reward is always positive for any action taken

Q5 The material describes supervised learning as: "We give a computer a bunch of choice questions (training samples) and provide standard answers."

In this analogy, what do "choice questions" and "standard answers" correspond to in a spam detection system?

A. Choice questions = emails; Standard answers = spam/ham labels
B. Choice questions = email features; Standard answers = email content
C. Choice questions = spam rules; Standard answers = classification accuracy
D. Choice questions = test emails; Standard answers = training emails

Q6 A Huawei autonomous vehicle encounters a flashing yellow traffic light. The vehicle must decide whether to brake or accelerate. The system learns from the outcome of each decision (safe passage vs. accident vs. traffic violation) and adjusts future behavior.

According to the material, this is BEST described as:

A. Supervised learning because the traffic rules are known
B. Unsupervised learning because the vehicle groups similar traffic situations
C. Reinforcement learning because the vehicle maximizes reward through environmental interaction
D. Semi-supervised learning because some traffic situations are labeled

Q7 The material mentions: "Machine learning evolution is producing new machine learning types, for example, self-supervised learning, contrastive learning, generative learning."

Which of these emerging types is MOST closely related to the semi-supervised learning paradigm described in the material?

A. Self-supervised learning — generates labels from the data itself
B. Contrastive learning — learns by comparing similar and dissimilar pairs
C. Generative learning — creates new data samples
D. All of the above are equally related

Multiple Choice Questions (8-11) ⚠ (Select ALL that apply)

Q8 Which of the following scenarios are VALID applications of unsupervised learning according to the material? (Select all that apply)

A. Grouping products with similar monthly sales patterns
B. Classifying emails as spam or ham with known labels
C. Recommending movies by grouping users with similar preferences
D. Defining fish species based on physical feature similarities without predefined categories
E. Predicting next week's stock price using historical data

Q9 The material states: "Both good and bad behavior can help reinforcement learning models learn."
Which statements about reinforcement learning are CORRECT based on the material? (Select all that apply)
A. The environment provides scalar reinforcement signals rather than correct actions
B. Positive rewards reinforce good behavior; negative rewards (penalties) discourage bad behavior
C. Reinforcement learning differs from supervised learning because it doesn't receive explicit correct answers
D. The model learns to maximize the cumulative reward over a series of actions
E. Reinforcement learning requires a fixed dataset with pre-labeled actions

Q10 A web page recommendation system has limited labeled data (few users label liked pages) and massive unlabeled data (all web pages). The material suggests using semi-supervised learning.

Which statements justify this choice? (Select all that apply)

A. Labeling data is labor-consuming and time-consuming
B. Unlabeled data is easily obtainable but labeled data is scarce
C. Semi-supervised learning can leverage both labeled and unlabeled data
D. Pure supervised learning would ignore valuable unlabeled data
E. Pure unsupervised learning would waste the available labeled information

Q11 Compare the four main types of machine learning described in the material. Which statements are ACCURATE? (Select all that apply)

A. Supervised learning requires labeled training data; unsupervised learning does not
B. Semi-supervised learning uses a small amount of labeled data + large amount of unlabeled data
C. Reinforcement learning learns from environmental feedback rather than static datasets
D. All four types can be used for classification tasks
E. Unsupervised learning always produces less accurate results than supervised learning

True or False Questions (12-15)

Q12 In supervised learning, the computer adjusts model parameters to make predictions close to standard answers, then can apply this to test samples without known answers.

A. True
B. False

Q13 The material states that unsupervised learning algorithms "do not know the answers to these questions, but it thinks that the answers to the questions in the same category should be the same." This means unsupervised learning can perform classification without any labels.

A. True
B. False

Q14 Reinforcement learning is essentially the same as supervised learning because both require knowing the correct action to learn effectively.

A. True
B. False

Q15 According to the material, clustering is a type of unsupervised learning that groups objects based on similarities, and can be used for applications like defining fish species and movie recommendations.

A. True
B. False

Fill in the Blank Questions (16-18)

Q16 In supervised learning, the program takes a known set of __________ and trains an optimal __________ to generate predictions. The trained model maps all inputs to __________ and performs simple judgment on the outputs.

Q17 Reinforcement learning differs from supervised learning in that, instead of telling the system the correct __________, the __________ provides scalar __________ signals to evaluate its actions.

Q18 Semi-supervised learning trains a model through a combination of a __________ amount of labeled data and a __________ amount of unlabeled data. This is particularly useful when labeling data is __________ and __________, but unlabeled data is easily obtainable.

Drag and Drop / Matching Questions (19-20)

Q19 Match each scenario with the correct machine learning type:

Scenario	**Learning Type**
A. Email spam detection with 10,000 labeled emails	**1. Supervised learning**
B. Customer segmentation without any labels	**2. Unsupervised learning**
C. Medical diagnosis with 500 labeled + 50,000 unlabeled X-rays	**3. Semi-supervised learning**
D. Robot learning to walk through trial and error	**4. Reinforcement learning**

Write your answer as: A→?, B→?, C→?, D→?

Q20 Match each characteristic to the correct learning type:

Characteristic	**Learning Type**
A. "Provides standard answers to choice questions"	**1. Supervised learning**
B. "Analyzes relationships and classifies without answers"	**2. Unsupervised learning**
C. "Combines limited labeled with massive unlabeled data"	**3. Semi-supervised learning**
D. "Maximizes reward function through environmental interaction"	**4. Reinforcement learning**

Write your answer as: A→?, B→?, C→?, D→?

🔑 Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	C	Semi-supervised learning is designed for small labeled + large unlabeled data
2	C	Action generation without knowing reward = unsupervised-like; Reward-based adjustment = supervised-like
3	B	Predicting exact price = continuous output = Supervised regression
4	B	Material explicitly states environment provides scalar reinforcement signals, not correct actions
5	A	Choice questions = input data (emails); Standard answers = labels (spam/ham)
6	C	Vehicle learns from environmental feedback (outcomes) to maximize safety reward = RL
7	A	Self-supervised learning generates pseudo-labels from data, similar to semi-supervised's label efficiency
8	A, C, D	B and E are supervised learning tasks
9	A, B, C, D	E is false - RL doesn't need pre-labeled datasets
10	A, B, C, D, E	All statements correctly justify semi-supervised learning for this scenario
11	A, B, C	D is false (RL is not typically for classification); E is false (unsupervised can be more suitable for some tasks)
12	A - True	Direct description from material
13	A - True	Unsupervised learning creates categories without labels, effectively performing classification-like grouping
14	B - False	Material explicitly states RL differs from supervised learning - doesn't receive correct actions
15	A - True	Direct quote from material's clustering section
16	samples, model, outputs	From supervised learning definition
17	action, environment, reinforcement	Fromf reinforcement learning vs supervised learning comparison
18	small, large, labor-consuming, time-consuming	From semi-supervised learning description
19	A→1, B→2, C→3, D→4	Direct matching of scenarios to types
20	A→1, B→2, C→3, D→4	Direct matching of characteristics to types

Module 2.3: Machine Learning Process — 20 Questions

Single Choice Questions (1-7)

Q1 A Huawei data science team is building a model to predict server failures in a data center. After collecting 6 months of logs, they discover that 40% of the temperature sensor readings are missing, 15% contain impossible values (e.g., -999°C), and the timestamp formats are inconsistent across different server racks.

According to the material's data processing framework, which step should they execute FIRST?

A. Feature extraction and selection
B. Data cleansing
C. Model training
D. Data preparation

Q2 The material shows that data scientists spend 60% of their time on "Cleansing and organizing data", while only 4% on "Optimizing models" and 3% on "Rebuilding training sets".

Which conclusion is MOST supported by this data?

A. Model selection is more important than data quality
B. Data quality directly determines model performance, making preprocessing the most critical phase
C. Most data scientists are inefficient and should automate data cleansing
D. The CrowdFlower report is outdated and no longer applicable

Q3 A dataset contains the following dirty data issues as shown in the material's "Dirty Data" table:

- **Row 3: Missing value in "City" column**
- **Row 5: Invalid value "A" in "Gender" column**
- **Row 8: Misfielded value "Itlay" instead of "Italy" in "Country"**
- **Row 10: Dependent attributes (IsTeacher=1 but #Students=26, which is inconsistent)**

Which data cleansing operation addresses the "misfielded value" problem?

A. Data filtering
B. Handling of possible error or abnormal values
C. Merging of data from multiple sources
D. Data loss handling

Q4 The material presents three feature selection methods: Filter, Wrapper, and Embedded.

A Huawei Cloud customer wants to build a spam detection model using 10,000 email features. They need to identify the most relevant features quickly without training multiple models, but they are concerned that the selected features might be redundant.

Which method should they choose, and what is the trade-off?

A. Filter method — fast but may select redundant variables
B. Wrapper method — computationally intensive but avoids redundancy
C. Embedded method (LASSO) — balances speed and redundancy but is model-specific
D. All methods are equally suitable for this scenario

Q5 According to the "Overall Process of Building an AI Model" diagram in the material, after splitting data into training, validation, and test sets, what is the CORRECT sequence of steps?

A. Train → Validate → Test → Deploy → Tune
B. Train → Validate → Test → Tune → Deploy
C. Train → Test → Validate → Deploy → Tune
D. Split → Train → Validate → Test → Tune → Deploy

Q6 A model is trained to predict house prices using features: Area, Location, Orientation. The training error is 5% but the test error is 35%. The material identifies this as a specific problem.

Which combination of solutions is MOST appropriate based on the material?

A. Use a more complex model + extend training time
B. Simplify the mathematical model + use dropout/weight decay methods
C. Increase training data + add more features
D. Reduce model capacity + decrease training data

Q7 The material shows the target diagram where:

- **Bias = difference between average predicted values and actual values**
- **Variance = how much predictions deviate from the mean**
A model consistently predicts house prices that are $50,000 lower than actual values across all test samples, but the predictions are very close to each other (low spread).

This model exhibits:

A. Low bias, low variance
B. Low bias, high variance
C. High bias, low variance
D. High bias, high variance

Multiple Choice Questions (8-11) ⚠ (Select ALL that apply)

Q8 The material states: "Data is crucial to models and determines the scope of model capabilities. All good models require good data."
Which data preprocessing operations are EXPLICITLY mentioned in the material as part of data preprocessing? (Select all that apply)
A. Data filtering
B. Data loss handling
C. Handling of possible error or abnormal values
D. Merging of data from multiple sources
E. Feature augmentation

Q9 The material describes three feature selection methods with their limitations:
Which statements about these methods are CORRECT? (Select all that apply)
A. Filter methods use statistics like Pearson correlation and Chi-square coefficient
B. Wrapper methods train a new model for each feature subset, making them computationally intensive
C. Embedded methods (like LASSO) treat feature selection as part of model training
D. Filter methods consider relationships between features and avoid redundancy
E. Wrapper methods usually provide high-performance feature sets for any type of model

Q10 The material discusses underfitting and overfitting with the complexity-error curve:
Which statements about model complexity and errors are CORRECT? (Select all that apply)
A. As model complexity increases, training error continuously decreases
B. Test error decreases initially then increases again, forming a convex curve
C. Underfitting occurs when model capacity is insufficient for task complexity
D. Overfitting occurs when training error is small but generalization error is large
E. The optimal model complexity is where training error equals test error

Q11 A Huawei image classification model is evaluated using 200 images (170 cats, 30 non-cats). The confusion matrix shows:

- **TP = 140, FP = 20, FN = 30, TN = 10**

Which statements are CORRECT? (Select all that apply)

A. Precision = 87.5%
B. Recall = 82.4%
C. Accuracy = 75%
D. The model has more false negatives than false positives
E. Specificity = 33.3%

True or False Questions (12-15)

Q12 In the machine learning process diagram, the feedback and iteration arrow goes from "Model deployment and integration" back to "Data preparation", indicating that deployed models may require retraining with new data.

A. True
B. False

Q13 The material states that "most machine learning models process features, which are usually numeric representations of input variables." Therefore, categorical data like "Orientation" (South, North, East) can be directly used without any conversion.

A. True
B. False

Q14 According to the material, the prediction error formula is: Prediction error = Bias² + Variance + Ineliminable error. This means that even with infinite data and a perfect model, some error will always remain due to the ineliminable error component.

A. True
B. False

Q15 The "What Is a Good Model?" slide identifies three factors: Generalization, Explainability, and Prediction speed. The material explicitly states that generalization capability is the most important factor among these three.

A. True
B. False

Fill in the Blank Questions (16-18)

Q16 In the machine learning process, the six main steps are: Data __________, Data __________, Feature __________ and __________, Model __________, Model __________, and Model __________ and __________.

Q17 The material identifies three types of "dirty data" problems: __________ (incomplete data or lack of attributes), __________ (incorrect records or abnormal values), and __________ (conflicting data).

Q18 In the bias-variance tradeoff, an ideal model should have __________ bias and __________ variance. However, the material states that it is impossible for a model to achieve both __________ (accurately capture training data rules) and __________ (generalize to new data) simultaneously.

Drag and Drop / Matching Questions (19-20)

Q19 (Match Error Type to Characteristic)

Match each error type to its characteristic based on the material:

Error Type	Characteristic
A. Underfitting	**1. Training error is small, generalization error is large**
B. Overfitting	**2. Training error is large, model is too simple**
C. Good fitting	**3. Low bias, low variance**
D. High model capacity	**4. Can handle complex tasks but may overfit**

Write your answer as: A→?, B→?, C→?, D→?

Q20 Match each data quality issue from the "Dirty Data" table to its example:

Data Issue	Example
A. Missing value	**1. "A" in Gender column**
B. Invalid value	**2. Empty City cell**
C. Misfielded value	**3. "Itlay" instead of "Italy"**
D. Incorrect format	**4. "1983-12-01" vs "31/12/1990" date formats**

Write your answer as: A→?, B→?, C→?, D→?

🔑 Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	B	Data cleansing is the first step after data preparation to handle missing/impossible/inconsistent values
2	B	The 60% vs 4% ratio emphasizes that data quality is the foundation of model performance
3	B	Misfielded values (wrong data in wrong field) are handled as "possible error or abnormal values"
4	A	Filter method is fastest (uses statistics only) but tends to select redundant variables as stated in material
5	B	Diagram shows: Split → Train → Validate → Test → Tune → Deploy (with feedback loop)
6	B	Small training error + large test error = overfitting. Material suggests: simplify model, use dropout/weight decay
7	C	Consistent \$50k underprediction = systematic error (high bias); close predictions = low variance
8	A, B, C, D	All four are explicitly listed in the data preprocessing section. E (feature augmentation) is mentioned under data conversion, not preprocessing
9	A, B, C	D is false (Filter methods DON'T consider feature relationships). E is false (Wrapper methods are specific to model type)
10	A, B, C, D	E is false - optimal point is where test error is minimized, not where training = test error
11	A, B, C, D	Precision = 140/160 = 87.5%; Recall = 140/170 = 82.4%; Accuracy = 150/200 = 75%; FN(30) > FP(20). E is false: Specificity = TN/N = 10/30 = 33.3% — actually TRUE! So A,B,C,D,E are all correct
12	A - True	The arrow clearly shows feedback from deployment back to data preparation
13	B - False	Material explicitly states categorical data needs encoding into numerals
14	B - False	Material states: "Theoretically, if there is infinite amount of data and a perfect model, the error can be eliminated"
15	A - True	Material explicitly states: "Generalization capability is the most important"
16	preparation, cleansing, extraction, selection, training, evaluation, deployment, integration	Six steps from the process diagram
17	Incompleteness, Noise, Inconsistency	Three types of dirty data from the material
18	low, low, accurately capture the rules in the training data, be generalized to invisible (new) data	From the bias-variance tradeoff section
19	A→2, B→1, C→3, D→4	Underfitting = simple model/large training error; Overfitting = small training error, large generalization error; Good fitting = low bias, low variance; High capacity = complex tasks but may overfit
20	A→2, B→1, C→3, D→4	Missing value = empty cell; Invalid value = "A" in Gender; Misfielded = "Itlay"; Incorrect format = date format mismatch

Module 2.4: Important Machine Learning Concepts — 20 Questions

Single Choice Questions (1-7)

Q1 A Huawei AI team is training a neural network with 10 million parameters on a dataset of 1 million images. The training is taking 48 hours per epoch using Batch Gradient Descent (BGD). The team needs to reduce training time while maintaining reasonable convergence stability.

Which approach should they choose based on the material's gradient descent comparison?

A. Continue with BGD but reduce the dataset size by 50%
B. Switch to Stochastic Gradient Descent (SGD) for faster iterations
C. Switch to Mini-batch Gradient Descent (MBGD) as a balance between speed and stability
D. Increase the learning rate by 10x to speed up convergence

Q2 The material shows that in gradient descent, "the gradient may fluctuate in a specific range of the target point" and SGD causes "instability" with "loss function fluctuation or even reverse displacement."

A model using SGD shows the following loss curve: rapid decrease for 100 iterations, then oscillation between 0.5 and 0.8 for the next 200 iterations without reaching the minimum.

What is the MOST likely cause according to the material?

A. The learning rate is too small
B. The learning rate is too large, causing the gradient to overshoot the minimum
C. The model is underfitting
D. The batch size is too large

Q3 In the hyperparameter analogy diagram, "Parameters are 'distilled' from data" while "Hyperparameters are used to control training."

Which of the following is a HYPERPARAMETER in a neural network?

A. Weight values between neurons
B. Bias terms in each layer
C. Learning rate
D. Output predictions

Q4 A data scientist is tuning a Support Vector Machine (SVM) model. They need to find the optimal values for C (regularization parameter) and σ (kernel parameter). The search space is large and they have limited computational budget.

According to the material, which hyperparameter search method is MOST appropriate?

A. Grid search with fine-grained steps for both parameters
B. Random search with a broad range first, then narrow based on best results
C. Manual tuning based on intuition
D. Bayesian search only, without any preliminary exploration

Q5 A Huawei medical AI team has a dataset of 500 patient records for a rare disease classification. They want to evaluate model performance robustly but are concerned about statistical uncertainty due to the small dataset.

Which approach does the material recommend?

A. Use a fixed 80/20 train-test split
B. Use 2-fold cross-validation to maximize training data
C. Use k-fold cross-validation with $k \geq 3$ (typically starting from 3)
D. Use the entire dataset for training and evaluate on the same data

Q6 A model is being trained with the following configuration:

- **Learning rate (η) = 0.1**
- **Batch size = 32**
- **Number of hidden layers = 3**
- **Number of neurons per layer = 128**
- **Activation function = ReLU**
- **L2 regularization parameter (λ) = 0.01**

Which of these are PARAMETERS and which are HYPERPARAMETERS?

A. All are parameters learned from data
B. Learning rate, batch size, layers, neurons, activation, λ are hyperparameters; weights/biases are parameters
C. Only weights are parameters; everything else is hyperparameters
D. λ is a parameter; everything else is a hyperparameter

Q7 The material states: "Grid search is expensive and time-consuming. This method works well when there are relatively few hyperparameters. Therefore, it is feasible for general machine learning algorithms, but not for neural networks."

Why is grid search NOT feasible for neural networks according to the material?

A. Neural networks don't have hyperparameters
B. Neural networks have too many hyperparameters (learning rate, batch size, activation function, number of neurons, number of iterations, etc.), making exhaustive search computationally prohibitive
C. Grid search only works for SVMs
D. Neural networks use random search exclusively by design

Multiple Choice Questions (8-11) ⚠ (Select ALL that apply)

Q8 The material compares three gradient descent methods (BGD, SGD, MBGD). Which statements are CORRECT? (Select all that apply)

A. BGD uses all training samples for each update and is the most stable
B. SGD uses one random sample per update and causes loss function fluctuation
C. MBGD uses a certain number of samples per update and balances BGD and SGD
D. BGD is the fastest method due to parallel processing of all samples
E. Reverse displacement in SGD is caused by noise data from random samples

Q9 The material distinguishes between parameters and hyperparameters. Which statements are CORRECT? (Select all that apply)

A. Parameters are automatically learned by models from training data
B. Hyperparameters are manually set by the user to control training
C. Weights in a neural network are parameters
D. Learning rate and number of iterations are hyperparameters
E. Support vectors in SVM are hyperparameters

Q10 The material discusses k-fold cross-validation with the following characteristics:

- **k is typically ≥ 2**
- **Usually starts from 3**
- **Set to 2 only when dataset is small**
- **Can effectively avoid over-learning and under-learning**

Which statements about k-fold CV are CORRECT? (Select all that apply)

A. k-fold CV splits data into k equal-sized subsets
B. Each subset serves as validation once while the remaining k-1 subsets serve as training
C. The average accuracy of k models is used as the performance metric
D. k in k-fold CV is itself a hyperparameter
E. When dataset has hundreds of thousands of samples, fixed train-test split is preferred over CV

Q11 The material describes a 4-step hyperparameter search process:

1. **Divide dataset into training, validation, and test sets**
2. **Optimize model parameters using training set**
3. **Search hyperparameters using validation set**
4. **Alternate steps 2 and 3 until convergence**

Which statements about this process are CORRECT? (Select all that apply)

A. The validation set is used for hyperparameter tuning, not for final model assessment
B. The test set should only be used once at the end to assess final performance
C. Grid search, random search, and Bayesian search are all valid search algorithms
D. Steps 2 and 3 are performed alternately until both parameters and hyperparameters are determined
E. The training set is used to evaluate the final model's generalization capability

True or False Questions (12-15)

Q12 In gradient descent, convergence means the objective function value changes very little or reaches the maximum number of iterations.
A. True
B. False

Q13 The material states that "some consider BGD and MBGD as types of SGD." This means BGD, SGD, and MBGD are fundamentally the same algorithm with only batch size differences.
A. True
B. False

Q14 According to the material, when the dataset is too small, alternative procedures (like k-fold CV) enable using all samples in the estimation of mean test error, at the price of increased computational cost.
A. True
B. False

Q15 Random search is more appropriate than grid search when the hyperparameter search space is large, because random search can find better solutions faster by sampling from possible values rather than exhaustively searching all combinations.

A. True
B. False

Fill in the Blank Questions (16-18)

Q16 The three main gradient descent methods are: __________ Gradient Descent (uses all samples), __________ Gradient Descent (uses one random sample), and __________ Gradient Descent (uses a subset of samples).

Q17 In the hyperparameter search process, step 1 divides the dataset into __________ set, __________ set, and __________ set. Step 2 optimizes __________ using the training set. Step 3 searches for __________ using the validation set.

Q18 The prediction error formula is: Prediction error = __________² + __________ + __________ error. The two main factors are __________ and __________. An ideal model should have __________ bias and __________ variance.

Drag and Drop / Matching Questions (19-20)

Q19 Match each gradient descent method to its characteristic:

Method	Characteristic
A. BGD	**1. Fastest but most unstable, causes fluctuation**
B. SGD	**2. Most stable but consumes too many compute resources**
C. MBGD	**3. Balance between stability and computational efficiency**

Write your answer as: A→?, B→?, C→?

Q20 Match each concept to its correct example from the material:

Concept	Example
A. Parameter	**1. Learning rate = 0.01**
B. Hyperparameter	**2. Weight between input and hidden layer = 0.35**
C. Grid search	**3. Exhaustive search of all combinations in a 4×4 grid**
D. Random search	**4. Sampling from possible values in a large search space**

Write your answer as: A→?, B→?, C→?, D→?

Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	C	MBGD balances speed (faster than BGD) and stability (more stable than SGD)
2	B	Material states SGD causes instability and fluctuation; large learning rate causes overshooting
3	C	Learning rate is manually set = hyperparameter; weights/biases are learned = parameters
4	B	Material recommends random search for large spaces: "first broad range, then narrow"
5	C	Material states k typically starts from 3; k=2 only when dataset is small; 500 records is small but k≥3 is still recommended
6	B	Learning rate, batch size, architecture, activation, λ are all hyperparameters; weights/biases are parameters
7	B	Neural networks have many hyperparameters (learning rate, batch size, activation, neurons, iterations), making grid search prohibitive
8	A, B, C, E	D is false - BGD is the slowest, not fastest, due to processing all samples
9	A, B, C, D	E is false - Support vectors are parameters learned from data, not hyperparameters
10	A, B, C, D	E is false - Material says when dataset is large, fixed split is "not a serious problem" but doesn't say it's preferred over CV
11	A, B, C, D	E is false - Training set is for training, not final evaluation; test set evaluates generalization
12	A - True	Direct definition from material
13	B - False	While related, they have fundamental differences in stability, convergence, and behavior
14	A - True	Direct quote from material's cross-validation section
15	A - True	Direct description from random search section
16	Batch, Stochastic, Mini-batch	Three methods from gradient descent section
17	training, validation, test, model parameters, hyperparameters	From the 4-step hyperparameter search process
18	Bias, Variance, Ineliminable, variance, bias, low, low	From prediction error formula and bias-variance tradeoff
19	A→2, B→1, C→3	BGD = most stable but resource-intensive; SGD = fastest but unstable; MBGD = balance
20	A→2, B→1, C→3, D→4	Parameters = learned values (weights); Hyperparameters = manually set (learning rate); Grid search = exhaustive; Random search = sampling

Module 2.5: Common Machine Learning Algorithms — 30 Questions

Single Choice Questions (1-10)

Q1 A Huawei real estate company wants to predict apartment prices based on area, layout, location, and floor level. The relationship between features and price is approximately linear, but with some non-linear interactions between area and location.

Which algorithm is MOST appropriate as a starting point?

A. Logistic regression
B. Linear regression
C. k-NN classification
D. Naive Bayes

Q2 The material states: "Polynomial regression is a type of linear regression. Although its features are non-linear, the relationship between its weight parameters w is still linear."

What does this statement fundamentally mean?

A. Polynomial regression uses a non-linear loss function
B. The model is linear in parameters (weights) even with non-linear features
C. Polynomial regression is actually a classification algorithm
D. The features themselves are linear transformations

Q3 A linear regression model is overfitting with very large weight values. The team wants to penalize large weights while keeping all features.

Which regularization should they use?

A. L1 regularization (Lasso) — tends to eliminate features
B. L2 regularization (Ridge) — keeps all features but penalizes large weights
C. No regularization — let the model fit perfectly
D. Dropout — only for neural networks

Q4 The material states: "Logistic regression and linear regression are both linear models in broad sense. The former introduces a non-linear factor (sigmoid function) on the basis of the latter and sets a threshold."

What is the PRIMARY purpose of the sigmoid function in logistic regression?

A. To make the model non-linear in parameters
B. To map the linear output to a probability value between 0 and 1
C. To increase model complexity for better fitting
D. To reduce the number of features

Q5 A Huawei image recognition system needs to classify fruits into 4 categories: Grape, Orange, Apple, Banana. The model outputs probability values for each class.

According to the material, which function should be used?

A. Sigmoid function
B. Softmax function
C. Linear activation
D. ReLU function

Q6 In the "Cheat" detection decision tree example from the material:

Refund = Yes → No Cheat
Refund = No → Check Marital Status
Marital Status = Married → No Cheat
Marital Status = Single/Divorced → Check Taxable Income
Taxable Income < 80K → No Cheat
Taxable Income > 80K → Yes Cheat
What is the root node, and why was it chosen first?

A. Marital Status — because it has the highest purity difference
B. Refund — because it provides the most significant initial split/purity difference
C. Taxable Income — because it's a continuous variable
D. Cheat — because it's the target variable

Q7 A dataset in 2D space has red and blue points arranged in concentric circles (red inside, blue outside). A linear SVM cannot separate them.

According to the material, what should be done?

A. Use a linear kernel with higher regularization
B. Use a non-linear kernel (e.g., Gaussian/RBF) to map to higher-dimensional space
C. Reduce the dataset size
D. Switch to k-NN algorithm

Q8 In the k-NN diagrams showing k=1, k=3, k=5, k=7, as k increases:
A. The decision boundary becomes more complex and overfits
B. The decision boundary becomes smoother, reducing overfitting risk
C. The model always achieves higher accuracy
D. The model requires less training data

Q9 The material states: "The Bayes classifier assumes that the effect of an attribute value on a given class is independent of the values of other attributes. This assumption is made to simplify the calculation and becomes 'naive' in this sense."
In a spam detection system using Naive Bayes, this means:
A. The presence of "free" is independent of "prize" in determining spam probability
B. The algorithm ignores all feature correlations
C. The model cannot handle multiple features
D. The algorithm requires infinite training data

Q10 The material describes ensemble learning: "If you ask thousands of people at random a complex question and then summarize their answers, the summarized answer is more accurate than an expert's answer in most cases."
Which ensemble method does this analogy BEST represent?
A. Boosting — sequential learning from mistakes
B. Bagging — independent learners combined by voting/averaging
C. Stacking — meta-learner combining different algorithms
D. Single decision tree

Multiple Choice Questions (11-16) ⚠ (Select ALL that apply)

Q11 According to the "Machine Learning Algorithm Overview" diagram in the material, which algorithms are listed under BOTH Classification AND Regression? (Select all that apply)
A. SVM
B. Neural network
C. Decision tree
D. Random forest
E. k-NN

Q12 The material describes linear regression with: $h(x) = w^T x + b$, and error ε ~ Normal distribution.
Which statements are CORRECT? (Select all that apply)
A. Simple linear regression has one independent variable; multiple has two or more
B. The loss function is derived using normal distribution and maximum likelihood estimation
C. Gradient descent is used to minimize the loss function and find optimal weights
D. In multi-dimensional space, the model is a hyperplane, not a straight line
E. Polynomial regression is NOT a type of linear regression

Q13 The material describes decision tree construction with feature selection, tree generation, and pruning.
Which statements are CORRECT? (Select all that apply)
A. Each non-leaf node denotes a test on an attribute
B. Purity is measured through information entropy and GINI coefficient
C. Common algorithms include ID3, C4.5, and CART
D. Pruning (pre-pruning and post-pruning) is used to reduce overfitting
E. All nodes except the root node are **called leaf nodes**

Q14 The material describes SVMs as "binary classification models" with "kernel trick."
Which statements are CORRECT? (Select all that apply)
A. SVMs maximize the margin between two categories
B. Support vectors are the data points closest to the decision boundary
C. The kernel trick maps linearly inseparable data to higher-dimensional space
D. Common kernels include Linear, Polynomial, Gaussian, and Sigmoid
E. SVMs can only handle binary classification, never multi-class

Q15 The material distinguishes between Bagging and Boosting:
Which statements are CORRECT? (Select all that apply)
A. Bagging independently builds multiple learners and averages predictions
B. Random forest is an example of Bagging
C. Boosting builds learners in sequence to gradually reduce bias
D. AdaBoost, GBDT, and XGBoost are examples of Boosting
E. Bagging reduces variance; Boosting may overfit due to strong fitting capability
Q16
The material states: "k-NN is a non-parametric method and is often used for datasets with irregular decision boundaries."
Which statements are CORRECT? (Select all that apply)
A. k-NN uses majority voting for classification and mean value for regression
B. A larger k reduces noise impact but makes boundaries less distinct
C. A small k increases overfitting probability; large k increases underfitting probability
D. k-NN requires a very large amount of computation
E. k-NN builds an explicit model during training

True or False Questions (17-20)

Q17 In linear regression, the model function $h(x) = w^T x + b$ is a straight line only when x is one-dimensional; it becomes a hyperplane when x is multi-dimensional.
A. True
B. False

Q18 Lasso regression uses L1-norm regularization (absolute value penalty), which tends to eliminate insignificant features by driving their weights to exactly zero.
A. True
B. False

Q19 The material states that in random forest, "Bootstrap sampling" is used to create subsets, multiple decision trees are built, and results are aggregated. For classification, majority voting is used; for regression, mean value is used.
A. True
B. False

Q20 In GBDT (Gradient Boosted Decision Tree), the next base learner tries to fit the residual (error) of the previous learner's prediction, and the final prediction is the sum of all base learners' results.
A. True
B. False

Fill in the Blank Questions (21-24)

Q21 In linear regression, the model function is h(x) = __________ + __________, where w is the __________ parameter and b is the __________.

Q22 The softmax function compresses a k-dimensional vector of arbitrary real values to another k-dimensional vector where each element is in __________. The sum of all probabilities equals __________. In the fruit classification example, the class with the highest probability was __________ with __________%.

Q23 SVMs find a __________ that maximizes the __________ between the two categories. The data points closest to this boundary are called __________. For linearly inseparable data, __________ functions are used to map data to a higher-dimensional __________ space.

Q24 In ensemble learning, __________ (example: random forest) independently builds multiple basic learners and averages their predictions, reducing __________. __________ (example: GBDT, XGBoost) builds learners in sequence, gradually reducing __________, but may cause __________.

Drag and Drop / Matching Questions (25-30)

Q25 Match each algorithm to its primary learning type:

Algorithm	Type
A. Linear regression	1. Classification
B. Logistic regression	2. Regression
C. k-means clustering	3. Clustering (Unsupervised)
D. Random forest	4. Ensemble

Write your answer as: A→?, B→?, C→?, D→?

Q26 Match each regularization to its property:

Regularization	Property
A. L1 (Lasso)	1. Square sum penalty, keeps all features
B. L2 (Ridge)	2. Absolute value penalty, may eliminate features
C. No regularization	3. May overfit with large weights

Write your answer as: A→?, B→?, C→?

Q27 Match each SVM kernel to its description:

Kernel	Description
A. Linear kernel	1. Most frequently used for non-linear data
B. Gaussian/RBF kernel	2. Simple, for linearly separable data
C. Polynomial kernel	3. Maps data using polynomial features

Write your answer as: A→?, B→?, C→?

Q28 Match each k-value behavior:

k-value	Behavior
A. Small k (e.g., k=1)	1. Higher probability of underfitting, rough division
B. Large k (e.g., k=15)	2. Higher probability of overfitting, sensitive to noise
C. Optimal k	3. Balance between bias and variance

Write your answer as: A→?, B→?, C→?

Q29 Match each ensemble type to its characteristic:

k-value	Behavior
A. Small k (e.g., k=1)	1. Higher probability of underfitting, rough division
B. Large k (e.g., k=15)	2. Higher probability of overfitting, sensitive to noise
C. Optimal k	3. Balance between bias and variance

Write your answer as: A→?, B→?, C→?

Q30 Match each Huawei Cloud customer scenario to the BEST algorithm:

Scenario	Algorithm
A. Predict exact server CPU usage for next 24 hours	1. Linear regression
B. Classify support tickets as "Urgent/Normal/Low"	2. Softmax regression
C. Group customers by purchasing behavior without labels	3. k-means clustering
D. Detect fraud with complex non-linear patterns	4. SVM with Gaussian kernel
E. Spam detection with word independence assumption	5. Naive Bayes

Write your answer as: A→?, B→?, C→?, D→?, E→?

🔑 Answer Key (Don't peek until you finish!)

Q	Answer	Explanation
1	B	Linear regression is the starting point for predicting continuous values with approximately linear relationships
2	B	"Linear in parameters" means weights w have linear relationship, even with polynomial features
3	B	L2 (Ridge) penalizes large weights but keeps all features; L1 (Lasso) eliminates features
4	B	Sigmoid maps (-∞, +∞) to (0, 1), producing probability for binary classification
5	B	Softmax generalizes logistic regression to multi-class (k-class) classification
6	B	Refund is the root node because it provides the most significant purity difference for initial split
7	B	Non-linear kernel (Gaussian/RBF) maps to higher-dimensional space where data becomes linearly separable
8	B	Material states: "boundary becomes smoother as k value increases"
9	A	Naive assumption: features are conditionally independent given the class
10	B	"Thousands of people at random" = independent learners = Bagging
11	A, B, C, D, E	All five algorithms appear in BOTH Classification and Regression branches of the overview diagram
12	A, B, C, D	E is false - material explicitly states "Polynomial regression is a type of linear regression"
13	A, B, C, D	E is false - "All nodes except the root node are called leaf" is incorrect; only terminal nodes are leaves
14	A, B, C, D	E is false - SVMs CAN handle multi-class using one-vs-one or one-vs-rest strategies
15	A, B, C, D, E	All statements correctly describe Bagging and Boosting from the material
16	A, B, C, D	E is false - k-NN is "lazy learning", no explicit model is built during training
17	A - True	Direct quote from material
18	A - True	L1 penalty drives some weights to exactly zero, effectively eliminating features
19	A - True	Direct description from random forest section
20	A - True	Direct description from GBDT section
21	$w^T x$, b, weight, bias	From linear regression model function
22	(0, 1), 1, Apple, 68	From softmax example: Apple=0.68, sum=1, each in (0,1)
23	hyperplane, margin, support vectors, kernel, feature	From SVM description
24	Bagging, variance, Boosting, bias, overfitting	From ensemble learning types comparison
25	A→2, B→1, C→3, D→4	Linear regression=Regression, Logistic=Classification, k-means=Clustering, RF=Ensemble
26	A→2, B→1, C→3	L1=absolute/eliminates features, L2=square/keeps features, None=may overfit
27	A→2, B→1, C→3	Linear=simple linear, Gaussian=most frequent non-linear, Polynomial=polynomial features
28	A→2, B→1, C→3	Small k=overfitting, Large k=underfitting, Optimal=balance
29	A→2, B→1, C→3	Bagging=independent/variance reduction, Boosting=sequential/bias reduction, RF=Bagging+DT
30	A→1, B→2, C→3, D→4, E→5	CPU=regression, Tickets=multi-class softmax, Customers=clustering, Fraud=non-linear SVM, Spam=Naive Bayes